

E-COMMERCE CUSTOMER & REVENUE DECISION INTELLIGENCE SYSTEM — ANALYSIS REPORT

Prepared by: Anannya Dutta

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1. Executive Summary

This report looks at the Olist e-commerce dataset that covers customer orders, payments, reviews, products and logistics from September 2016 to July 2018. It checks what drives revenue, how people pay, how products perform in categories and how fast deliveries happen.

- The study looks at about 93,000 orders and \$14.98 million in revenue that was tracked.
- It finds that most demand comes from the São Paulo region.
- It also shows that a single payment method, credit card makes up 78% of revenue.
- The average delivery time is 28% longer than the target of ten days.
- These findings help recommend ways to improve logistics, spread payment risk and invest in product categories.

2. Project Introduction

2.1 Project Title

E-Commerce Customer & Revenue Decision Intelligence System

2.2 Objective

The objective of this project was to build an end-to-end analytical system — from relational data modeling through to an interactive Power BI dashboard — that answers core business questions about revenue trends, customer payment behavior, product category performance, and logistics/delivery efficiency for an e-commerce marketplace, enabling data-driven decisions on where to invest and where to intervene operationally.

2.3 Dataset Overview

- Source: Olist Brazilian E-Commerce Public Dataset (Kaggle).
- Scope: ~93,000 unique orders covering transactions from September 2016 through July 2018.
- Key Tables: customers, orders, order_items, order_payments, order_reviews, products, sellers, geolocation, and product_category_name_translation.

- Key Attributes: Order ID, Customer ID, order/delivery timestamps, product category, price, freight value, payment type, and review score.

2.4 Tools Used

- MySQL: Schema design, relational joins, and KPI aggregation queries.
- Python (Pandas, NumPy): Data cleaning and exploratory data analysis.
- Power BI: Data modeling (star schema), DAX measures, and interactive dashboard.

3. Methodology

3.1 Data Inspection

- Reviewed all 9 source tables for structure, keys, and business meaning.
- Identified relationships between orders, customers, products, sellers, payments and reviews.

3.2 Data Validation

- Checked data types across columns (e.g., price and freight as numeric, order/delivery dates as datetime).
- Validated referential integrity between orders and related tables (payments, reviews, order items).
- Confirmed primary key uniqueness across dimension and fact tables.

3.3 Data Cleaning Strategy

- Handled missing attributes and standardized date/timestamp formats across order purchase, shipping, and delivery fields.
- Identified and removed trailing incomplete month records (August 2018) to prevent artificial volume drop-off distortions, establishing a clean analytical boundary from September 2016 through July 2018.
- Standardized product category translations to ensure clean grouping and reporting.

3.4 Data Aggregation

- Revenue trend aggregation by year, quarter and month
- Category-level revenue and freight cost aggregation
- Payment-type revenue aggregation
- Order volume and revenue aggregation by customer state
- Delivery gap calculation (actual delivery date minus estimated delivery date)

3.5 KPI Calculation

KPI	Value
Total Revenue	\$14.98 M
Total Orders	93 K
Average Order Value	\$161.25
Total Freight Cost	\$2.10 M (14.1% Revenue)
Average Review Score	4.07 / 5.0
Average Delivery Days	12.8 days (Target: 10 days)

4. Exploratory Analysis

4.1 Revenue by Product Category

Category	Revenue
health_beauty	\$1.14 M
bed_bath_table	\$0.98 M
sports_leisure	\$0.94 M
computers_accessories	\$0.87 M
furniture_decor	\$0.69 M

Top five product categories drive over \$4.6M in sales, led by health & beauty at \$1.14M.

4.2 Revenue by Payment Type

Payment Type	Revenue
credit_card	\$11.70 M
boleto	\$2.70 M
voucher	\$0.30 M
debit_card	\$0.20 M

Credit cards account for 78.3% of total revenue, representing significant single-channel payment reliance.

4.3 Orders by Customer State

State	Order Volume
SP (São Paulo)	38 K
RJ (Rio de Janeiro)	12 K
MG (Minas Gerais)	11 K
RS (Rio Grande do Sul)	5 K
PR (Paraná)	5 K

São Paulo dominates platform order volume with 38K orders—more than triple the second-largest state.

4.4 Freight Cost by Product Category

Category	Freight Cost
bed_bath_table	\$191 K
health_beauty	\$166 K
furniture_decor	\$162 K
sports_leisure	\$160 K
computers_accessories	\$139 K

Bed, bath, and table goods generate the highest total freight expense (\$191K), outpacing top-revenue categories due to product dimensions and shipping weight profiles.

5. Dashboard Development

Key Visual Components:

- Executive KPI Cards: Top-level summary metrics tracking Total Revenue (\$14.98M), Total Orders (93K), Average Order Value (\$161.25), Total Freight Cost (\$2.10M / 14.1%), Average Review Score (4.07), and Average Delivery Days (12.8d).
- Temporal Revenue Trajectory: Dynamic line visual tracking monthly and quarterly sales performance with synchronized date slicers.
- Category Performance & Freight Breakdown: Bar charts evaluating gross sales alongside freight costs by product category.
- Payment Method Breakdown: Horizontal bar chart highlighting payment modality breakdown and credit card volume dominance (\$11.7M).
- Geographic Order Concentration: Horizontal bar chart displaying state-level order volume led by São Paulo (38K).

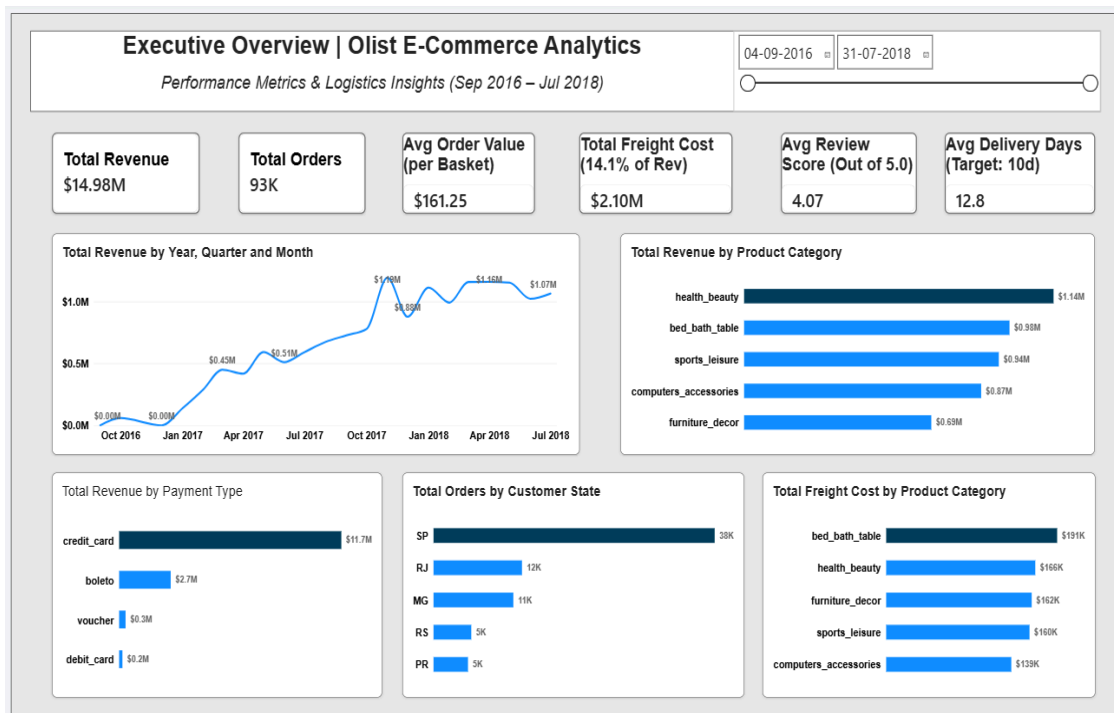


Figure 1: Interactive Power BI Executive Dashboard displaying key revenue metrics, temporal trends, category breakdowns, and logistics performance (Sep 2016 – Jul 2018).

6. Key Insights

6.1 Revenue Concentration in Top Product Categories

- health_beauty leads all categories at \$1.14M in revenue, with the top five categories combined generating over \$4.6M.
- This concentration suggests that prioritized marketing campaigns and inventory allocation toward these specific core categories could yield higher overall returns than broader, catalog-wide investments.

6.2 Single-Channel Payment Dependency Risk

- Credit cards account for 78.3% of total tracked revenue (\$11.7M out of \$14.98M).
- While convenient for consumers, this heavy dependence leads to one-way risk — which means if a payment processor goes down or if the costs to process payments change or if new rules come out it would directly impact the vast majority of sales.

6.3 Regional Demand Dominance

- São Paulo (SP) represents 38K orders, which is more than three times the volume of the next-highest state, Rio de Janeiro (12K orders).
- This indicates strong logistics maturity and customer adoption in SP, while highlighting potential growth opportunities in under-penetrated regional markets across Brazil.

6.4 Operational SLA Delivery Gap

- The average delivery lead time stands at 12.8 days against the internal target SLA of 10 days—a 28% variance.
- Although the platform maintains a solid overall review score of 4.07 out of 5.0, this fulfillment lag suggests operational friction in last-mile logistics that warrants further investigation.

6.5 Disconnect Between Freight Expense and Category Sales

- bed_bath_table generates the highest overall freight expenditure (\$191K) despite ranking second in overall revenue, whereas top-revenue category health_beauty incurs lower shipping costs (\$166K).
- This pattern suggests that product dimensions, package weight, and shipping efficiency vary significantly by category.

7. Growth Opportunities & Recommendations

7.1 Category Merchandising & Inventory Expansion (Insight 6.1)

Prioritize promotional campaigns, featured homepage placements, and seller acquisition toward health_beauty and the top 5 revenue categories to maximize returns on marketing spend.

7.2 Payment Channel Diversification (Insight 6.2)

Introduce targeted incentives (e.g., small discounts or checkout promos) for boleto or debit payments to reduce single-processor exposure and hedge against credit card transaction fee risks.

7.3 Regional Market Penetration (Insight 6.3)

Launch localized marketing and regional seller onboarding programs in secondary markets like Rio de Janeiro (12K orders) and Minas Gerais (11K orders) to build market share beyond São Paulo.

7.4 Logistics SLA Bottleneck Audit (Insight 6.4)

Investigate root causes along carrier routes exceeding the 10-day delivery SLA (focusing on dispatch delays vs last-mile transit) to reduce the 12.8-day average lead time.

7.5 Freight Cost Optimization (Insight 6.5)

Conduct a packaging audit and renegotiate carrier rates specifically for bulkier categories like bed_bath_table to bring freight expenses (\$191K) in line with item margins.

8. Limitations

This analysis is subject to several analytical and structural boundaries that should be considered when interpreting the results:

Fixed Timeframe Boundary

The dataset covers transactions from September 2016 through July 2018 across the Brazilian marketplace, meaning findings may not reflect post-2018 consumer shifts, economic changes, or post-pandemic e-commerce adoption patterns.

Missing Dimensional Attributes

The absence of explicit product dimensions, package volumes, and unit shipping weights limits freight cost evaluations, preventing a full determination of whether higher freight costs in categories like bed_bath_table stem from physical bulk or inefficient carrier pricing.

Associative Performance Relationships

While logistics lead times exceed targets (12.8 days vs 10 days) alongside an average review score of 4.07, this relationship is purely associative; formal statistical testing (e.g., correlation or regression modeling) is required before establishing direct causality between delivery speed and customer satisfaction scores.

9. Appendix

9.1 Technical Implementation Details

The relational data schema was modeled in Power BI using a star-schema design, connecting fact tables (orders, order_items, order_payments, order_reviews) to core dimension tables (customers, products, sellers, geolocation).

9.2 Core SQL Analytics Queries

1. Revenue & Order Volume Aggregation (Sep 2016 – Jul 2018)

```
SELECT
  DATE_FORMAT(o.order_purchase_timestamp, '%Y-%m') AS order_month,
  COUNT(DISTINCT o.order_id) AS total_orders,
  ROUND(SUM(p.payment_value), 2) AS total_revenue
FROM orders o
JOIN order_payments p ON o.order_id = p.order_id
WHERE o.order_purchase_timestamp >= '2016-09-01'
  AND o.order_purchase_timestamp <= '2018-07-31'
GROUP BY order_month
ORDER BY order_month ASC;
```

2. Category Revenue & Freight Breakdown

```
SELECT
  t.product_category_name_english AS category_name,
  ROUND(SUM(i.price), 2) AS gross_revenue,
  ROUND(SUM(i.freight_value), 2) AS total_freight_cost
FROM order_items i
JOIN products p ON i.product_id = p.product_id
JOIN product_category_name_translation t
  ON p.product_category_name = t.product_category_name
GROUP BY t.product_category_name_english
ORDER BY gross_revenue DESC;
```

9.3 Power BI DAX Measure Engine

```
// Total Revenue
Total Revenue = SUM(order_payments[payment_value])

// Average Order Value (AOV)
AOV = DIVIDE([Total Revenue], DISTINCTCOUNT(orders[order_id]), 0)

// Total Freight Expense
Total Freight Cost = SUM(order_items[freight_value])

// Freight Cost Ratio
Freight % of Revenue = DIVIDE([Total Freight Cost], [Total Revenue], 0)

// Average Delivery Days
Avg Delivery Days =
AVERAGEX(
  FILTER(orders, NOT ISBLANK(orders[order_delivered_customer_date])),
  DATEDIFF(orders[order_purchase_timestamp],
    orders[order_delivered_customer_date], DAY)
)

// Average Review Score
Avg Review Score = AVERAGE(order_reviews[review_score])
```

9.4 Full Source Code & Project Repository

- GitHub Repository: <https://github.com/A12Dutta/ecommerce-revenue-intelligence>